Lesson 1: Spatial Puzzles: Tangrams and Pentominoes

Choosing a strategy, building shapes from the seven tangram pieces, finding all twelve pentominoes, and testing which ones fold into a box or fill a rectangle.

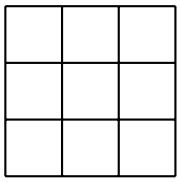
1. Strategies for Spatial Puzzles

A **spatial puzzle** is a puzzle about shapes and space: fitting pieces together, turning them, flipping them or folding them. A **strategy** is a plan for solving a problem. Good puzzle solvers know several strategies and pick the one that fits.

Problem-solving strategies	
Draw or model	Sketch the puzzle, or use real pieces that you can move around.
Guess and check	Try something. If it does not work, change it a little and try again.
Make a systematic list	List every case in an order, so you do not miss one or count one twice.
Look for a pattern	Solve small cases, see how the answers grow, and use that to predict.
Simplify the problem	Solve a smaller or easier version first.
Eliminate possibilities	Rule out the choices that cannot work, so fewer are left to try.
Work backward	Start from the end and undo the steps (you will use this for games in Lesson 2).
Try another approach	Solve it a second way to check (**verify**) your answer.
Four steps for any puzzle: 1. Understand the rules. 2. Plan: pick a strategy. 3. Solve. 4. Check that your answer follows the rules.	

Example 1

How many squares of *any* size are in this 3 by 3 grid?



(a) Make a systematic list by the size of the square.

Size of square	1 by 1	2 by 2	3 by 3	**Total**
How many				

(b) Simplify the problem: how many squares are in a 2 by 2 grid? Does the list in (a) look like it follows a pattern?

(c) Which strategies did you use?

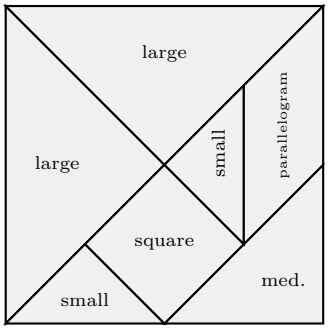
Watch out. Most people answer 9 because they only count the small squares. The bigger squares overlap the small ones, and they count too. A systematic list makes sure you get every size.

2. The Tangram

A **tangram** is a very old puzzle from China. A square is cut into 7 pieces: **2 large triangles**, **1 medium triangle**, **2 small triangles**, **1 square** and **1 parallelogram**. (A **parallelogram** is a four-sided shape whose opposite sides are parallel, like a slanted rectangle.) The goal is to use the pieces to make a given shape. Its outline, filled in, is called a **silhouette**.

Tangram rules: Lay the pieces flat. Pieces touch but do not overlap. You may turn and flip any piece. Use exactly the pieces the puzzle asks for.

Example 2



Use a **small triangle** as 1 unit of area. Imagine covering each piece with small triangles (draw them on the picture).

(a) Complete the table.

Piece	How many in the set	Area of one (small triangles)	Fraction of the whole square
large triangle	2		
medium triangle	1		
small triangle	2	1	
square	1		
parallelogram	1		

(b) What is the area of the whole square, in small triangles? Check by adding all 7 pieces.

(c) A tangram is cut from a square of card 20 cm by 20 cm. Find the area of each kind of piece.

Watch out. The square, the medium triangle and the parallelogram look very different, but they have the **same area** (2 small triangles each). Same area does not mean same shape.

3. Making Shapes with Tangram Pieces

Use your cut-out pieces (the cut-out page is at the end of this lesson). These tips use the strategies from Section 1.

Tangram tips
1. Check the area first (eliminate possibilities). The pieces you use must add up to the area of the silhouette.
2. Place the biggest pieces first. They fit in the fewest places, so there are fewer guesses to try.
3. Look at the corners. A square corner (90°) of the silhouette needs the square corner of a triangle, a corner of the square piece, or two sharp (45°) corners side by side.
4. Guess and check. If a gap is left that no piece fits, move one piece and try again. Remember you may flip pieces.

Example 3

Use only the **two small triangles**. Make each shape and sketch how the two triangles fit.

(a) a square

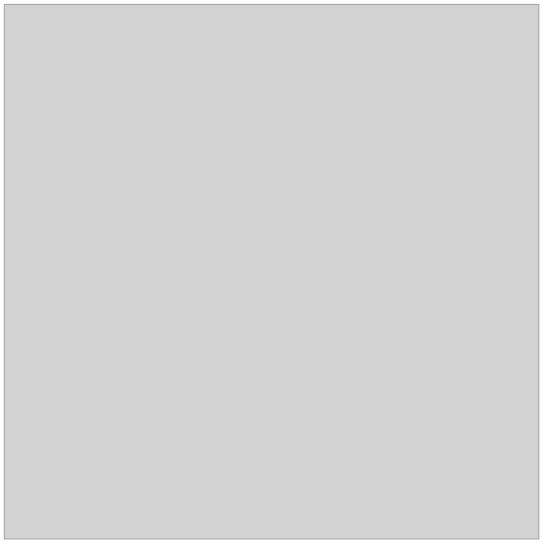
(b) a bigger triangle

(c) a parallelogram

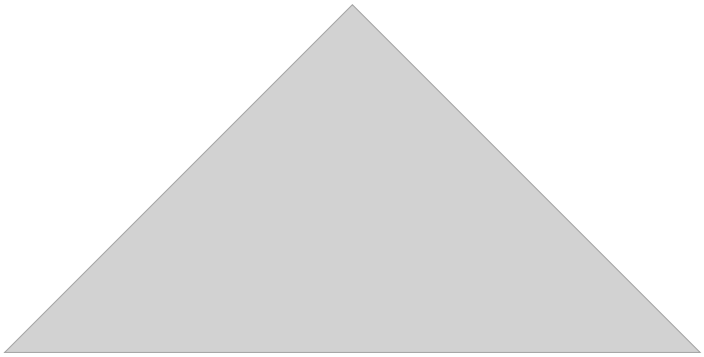
(d) Each shape you made has the same size and shape as one of the other tangram pieces. Which piece matches each one?

Example 4

(a) Make this square with 5 pieces: both small triangles, the medium triangle, the square and the parallelogram. The picture is the same size as your pieces, so you can build it on top. Then draw the lines.



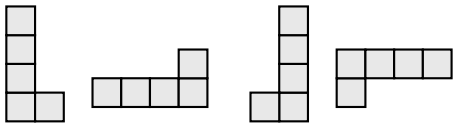
(b) Make this triangle with **all 7** pieces. (This picture is smaller than your pieces.) Draw the lines.



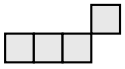
4. Pentominoes: Finding All 12

A **polyomino** is a shape made of equal squares joined **along full sides**. A domino has 2 squares. A shape made of **5** squares is a **pentomino**.

Two pentominoes are the **same** if one can be turned (**rotated**) or flipped over (**reflected**) to fit exactly on top of the other. So all four of these are the same pentomino:



the same pentomino, turned and flipped



not allowed: corners only

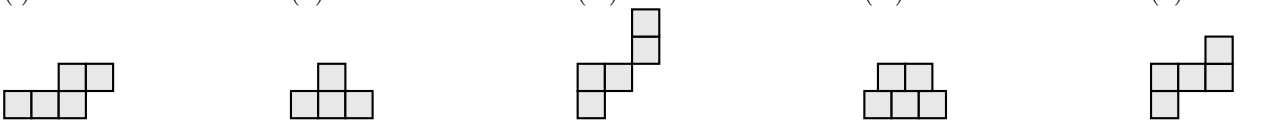


not allowed: half sides

Example 5

(a) Is each shape a pentomino? Write yes or no. If no, say why.

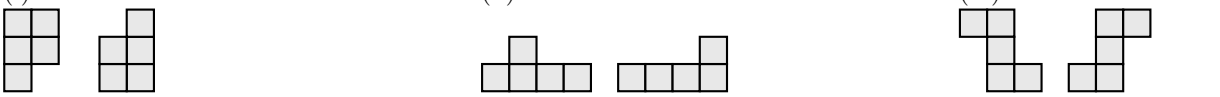
(i) (ii) (iii) (iv) (v)



_____ _____ _____ _____ _____

(b) Is each pair the same pentomino or different? If they are the same, say whether you turn or flip it.

(i) (ii) (iii)



_____ _____ _____

Example 6

Find all the different pentominoes with a **systematic list**.

Simplify first. With 3 squares there are only 2 shapes: 3 in a row, or a bent one. (With 4 squares there are 5. You will find them in the assignment.)

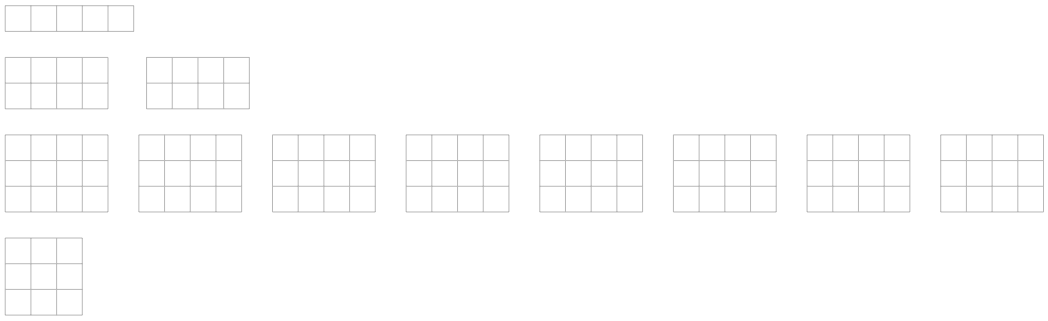
Sort by the longest straight row. Start with all 5 in a row, then 4 in a row, then 3, then 2. Before you draw a new one, check that it is not a turn or flip of one you already have. Draw each one on a grid below. The number of grids tells you how many to find.

5 in a row

4 in a row

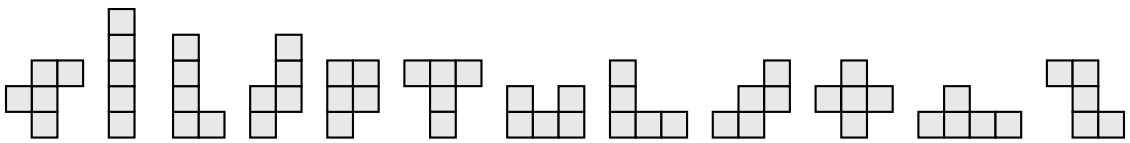
3 in a row

2 in a row



How many pentominoes are there in all? _____

Each pentomino is named by a capital letter that it looks a bit like. Draw your 12 on the grid paper on the cut-out page, write their letters, and cut them out.



F **I** **L** **N** **P** **T** **U** **V** **W** **X** **Y** **Z**

5. Pentomino Puzzles: Open Boxes and Rectangles

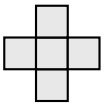
An **open box** is a cube with no top. It has 5 faces, so a pentomino might be a **net** for it (a flat pattern that folds up into the box, as in Unit 3, Lesson 10). But not every pentomino works.

Will it fold into an open box? (draw or model)
Step 1. Pick one square to be the bottom. **Step 2.** Fold the squares that touch it straight up: they are sides.
Step 3. Fold any other squares up against the sides. **Step 4.** If two squares land on the same face, it does not work. Try a different bottom square before you give up.
Quick eliminations: a 2 by 2 block of squares never folds (four squares meet at one point, but only three faces of a cube meet at a corner). Five in a row wraps all the way around and overlaps.

Example 7

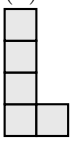
Does each pentomino fold into an open box? Write yes or no. If yes, shade the square that becomes the bottom. Check with your cut-out pieces.

(a) X



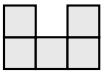
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(b) L



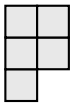
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(c) U



_____

(d) P



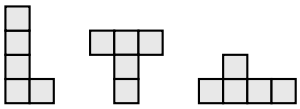
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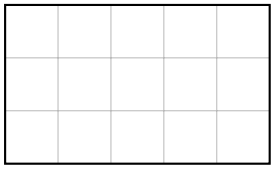
Tiling. To **tile** a rectangle means to cover it completely with pieces, with no gaps and no overlaps. Pieces may be turned and flipped.

Tiling tips **1.** Check the area: each pentomino covers 5 squares. **2.** Fill a corner or an edge first, where pieces fit in fewer ways. **3.** Eliminate: if you close off a gap whose number of squares is not a multiple of 5 (5, 10, 15, ...), that try cannot work. Undo the last piece and try again.

Example 8

Tile this 3 by 5 rectangle with the **L**, **T** and **Y** pentominoes. Draw the pieces on the rectangle and label them.



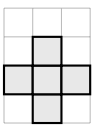


Example 9

Owen says: "The P pentomino has 5 squares and an open box has 5 faces, so P must fold into an open box."

(a) Identify and correct Owen's error.

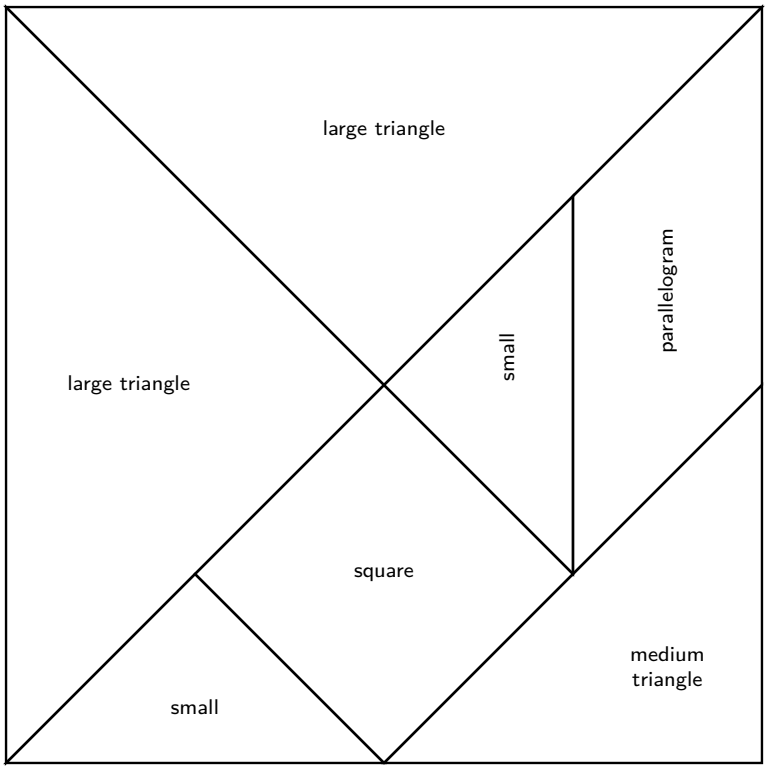
(b) Create a variation: a **closed** box (a cube) has 6 faces. Add one square to the X pentomino to make a net of a closed cube. Describe your strategy.



This page is blank on purpose. It is the back of the cut-out page, so you can cut that page up.

Cut-Out Page: Tangram and Pentomino Grid Paper

Tangram. Cut out the big square. Then cut along every line inside it to get the 7 pieces. Keep your pieces in an envelope.



Pentomino grid paper. After Example 6, draw each of the 12 pentominoes on this grid. Trace around each one darkly and write its letter on it. Then cut them out. (Each square is 1.5 cm. You can also fold them to test the open-box nets.)

Lesson 2: Spatial Games and Strategies

Playing simple games on piles, boards and grids, finding a strategy that wins, explaining why it works, fixing a strategy that does not, and changing the rules.

1. Take-Away Games: Work Backward

A **strategy** is a plan for solving a puzzle or winning a game. Good players do not just hope. They use problem-solving strategies like the ones in this toolbox. You will use all of them in this lesson.

Strategy toolbox

Guess and check: try something, see what happens, then adjust.

Look for a pattern: solve small cases and see what repeats.

Make a systematic list: list every possibility in an order, so none is missed or counted twice.

Draw or model: use a sketch, coins, counters or a real board.

Eliminate possibilities: cross out what cannot work.

Simplify the problem: try a smaller version first.

Work backward: start at the end (the win) and go back one step at a time.

Try another approach: if one way gets stuck, use a different strategy.

The take-away game. Put a pile of counters (coins, paper clips, beans) on the table. Two players take turns. On your turn you must take **1, 2 or 3** counters. The player who takes the **last** counter wins.

Example 1

Ana and Ben play with 10 counters. Ana goes first. Here is their game.

Turn	Player	Takes	Left
1	Ana	2	8
2	Ben	3	5
3	Ana	1	4
4	Ben	2	2
5	Ana	2	0

(a) Who won? _____

(b) At turn 4, Ben had 4 counters in front of him. Show that Ben could **not** win from there, whatever he took.

(c) **Work backward.** Start with small piles. For each number of counters in front of you, decide: can you win (W) or will you lose (L) if the other player plays well? If you can win, write how many to take.

Counters on your turn	1	2	3	4	5	6	7	8	9	10	11	12
Win or lose?												
Take												

(d) Finish the sentence. The losing numbers are _____, the _____.

Winning strategy (take 1, 2 or 3; last counter wins): always **leave a multiple of 4** (4, 8, 12, 16, 20, ...) for the other player.

How many to take: count down from the pile to the next multiple of 4 below it. (21 counters: the next multiple of 4 below is 20, so take 1.)

If the pile is already a multiple of 4 on your turn, you cannot force a win. Take 1 and hope the other player makes a mistake.

Example 2

You play the take-away game (take 1, 2 or 3, last counter wins) against a friend.

(a) The pile has 21 counters and you go first. Use the strategy to fill in the game.

Who	You	Friend	You	Friend	You	Friend
Takes		3		2		1
Left						

(b) With 11 left, how many do you take? Explain why you are sure to win from here.

(c) A new game starts with 24 counters. Should you go first or second? Explain.

Example 3

Change a rule, and fix a strategy.

(a) **Variation 1:** you may take 1, 2, 3 **or 4** counters. Last counter still wins. Which numbers should you leave now? How many should you take first from a pile of 23?

(b) **Variation 2:** take 1, 2 or 3, but the player who takes the last counter **loses**. Work backward. Which numbers should you leave? How many should you take first from 15?

(c) Jordan says, "In the 1, 2 or 3 game (last counter wins), always take 3. It makes the pile go down fastest." Show that Jordan's strategy can lose when there are 6 counters on his turn. What should he do instead?

2. Tic-Tac-Toe: Win, Block, Fork

Two players take turns on a 3×3 grid. X goes first. Each player puts a mark in an empty square. Three of your marks in a row (across, up and down, or diagonally) wins. If the grid fills up with no three in a row, the game is a **tie** (a draw). In this lesson the squares are numbered 1 to 9, like this:

1	2	3
4	5	6
7	8	9

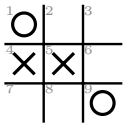
A **fork** is a move that makes **two** threats (two in a row with the third square empty) at the same time. The other player can block only one of them, so you win next turn.

Tic-tac-toe checklist (check every empty square, in this order, before you move):
1. Win. If you have two in a row and the third square is empty, take it.
2. Block. If the other player has two in a row and the third square is empty, take it.
3. Fork. If one move gives you two threats at once, take it.
4. Otherwise take the centre; if it is taken, take a corner. (One exception: if you are O and X holds two *opposite* corners, take an edge, or X can fork.)

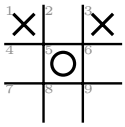
Example 4

Use the checklist. Where should the player whose turn it is play? Give the square number and the reason.

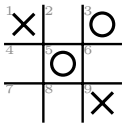
(a) X to play.



(b) O to play.

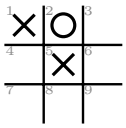


(c) X to play.



Example 5

Why good players always tie. Mateo (X) plays the centre. Sofia (O) answers on an edge, square 2. Mateo plays 1.



(a) Why must Sofia play 9 now?

(b) Mateo then plays 7. Show that he will win on his next turn.

(c) Sofia's first move was the mistake. People have checked every possible game (a systematic list): after X takes the centre, O must answer in a **corner**. Then, if both players use the checklist, the game always ends in a tie. Priya says, "If I go first and take the centre, I will always win." Correct her statement.

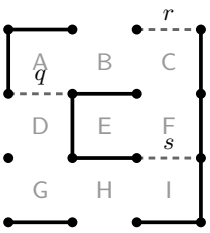
3. Dots and Boxes: Don't Draw the Third Side

Start with a grid of dots. Players take turns drawing one short line that joins two dots next to each other (across or up and down). If your line finishes the **fourth** side of a box, write your initial in the box and **take another turn**. When every line is drawn, the player with more boxes wins.

Dots and boxes strategy
1. If a box has 3 sides, take it (draw the fourth side).
2. Never draw the **third** side of a box if you can help it: the other player will take that box and move again.
3. Near the end, every line may give boxes away. Then give away the **smallest** group of boxes.

Example 6

It is your turn. The boxes are named A to I. The dashed lines q, r and s have **not** been drawn yet.



(a) Which box can you take right now? Which line do you draw?

(b) Now you must draw one more line. For each of q, r and s, say whether it is safe or which boxes it gives away.

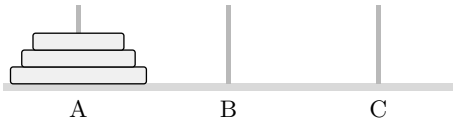
(c) Leo says, "I like to draw a side on a box that already has two sides. Then I'm closer to finishing it." What is wrong with Leo's strategy?

4. The Tower Puzzle: Simplify and Look for a Pattern

The **tower puzzle** has three pegs, A, B and C, and some discs of different sizes stacked on peg A, biggest at the bottom. The goal is to move the whole tower to peg C.

Rules
1. Move one disc at a time: the top disc of a peg.
2. Never put a disc on top of a smaller disc.

Model it with coins: a toonie, a loonie, a quarter, a nickel and a dime are five different sizes (biggest to smallest).



Example 7

What is the fewest number of moves for the tower puzzle? **Simplify the problem**: start with 1 disc, then 2, then 3.

(a) How many moves does 1 disc take? List the moves for 2 discs (S = small, L = large).

(b) Model 3 discs (S, M, L) and list the 7 moves. The first is done.

Move	1	2	3	4	5	6	7
Disc	S						
From	A						
To	C						

(c) Look at moves 1 to 3 in your table: they move the top _____ discs out of the way (onto B). Move 4 moves the _____ disc. Moves 5 to 7 move the 2 discs back on top. So for 4 discs: move the top 3 off, move the largest, move the 3 back on. Complete the table.

Discs	1	2	3	4	5	6
Fewest moves	1	3	7			

Tower puzzle: fewest moves for n discs $= 2 \times$ (moves for $n-1$ discs) $+ 1 \;=\; 2^n - 1$
Why: move the smaller discs out of the way, move the biggest disc once, then move the smaller discs back on top of it.

Calculator. For 8 discs: 2 [^] 8 [−] 1 [ENTER] shows 255.

Example 8

(a) A science centre has a tower puzzle with 8 discs. A visitor makes 1 move every second and never makes a mistake. How long does the puzzle take? Give minutes and seconds.

(b) Kai says, "1 disc takes 1 move and 2 discs take 3 moves, so each disc adds 2 moves. 5 discs take 9 moves." Find his error and correct it.

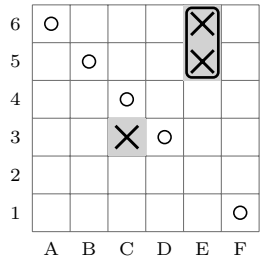
(c) **Variation:** a disc may move only to a peg **next to** its peg (A to B, B to C, C to B, B to A, but never straight between A and C). Model the puzzle. How many moves do 1 disc and 2 discs take? Does the rule $2^n - 1$ still work?

5. A Hidden-Ships Grid Game: Eliminate Possibilities

Each player has a secret 6×6 grid, with columns A to F across and rows 1 to 6 up. A square is named by its column, then its row, like a map: C4. Each player hides a **2-ship** (2 squares in a line) and a **3-ship** (3 squares in a line), across or up and down, never diagonal. Players take turns calling a square. The other player answers "hit" or "miss", and "sunk" when every square of a ship has been hit. The first player to sink all the other player's ships wins.

Hidden-ships strategy
Hunt: call only the squares of one colour in a checkerboard pattern. Two squares side by side always have different colours, so every ship covers at least one shaded square. You need at most half the squares (18 of the 36).
Target: after a hit, call the squares next to it (up, down, left, right). After two hits in a line, keep going along that line.
Eliminate: cross off any position that would use a square you know is a miss.

Example 9



X = hit, ∘ = miss. The 2-ship at E5–E6 is sunk.

You hit the 3-ship at C3. Your next two calls, C4 and D3, were misses.

(a) **Make a systematic list** of every place a 3-ship that covers C3 could be: 3 across, then 3 up and down. Then cross out the ones that use a miss.

(b) Which square should you call next? What will each answer tell you?

(c) Mia says, "The ship could still be C3, D3 and E3." What is her error?

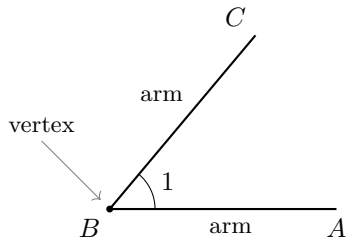
Lesson 3: Measuring, Drawing and Classifying Angles

Naming the parts of an angle, sorting angles by size, measuring and drawing angles with a protractor (even when the angle is turned), finding reflex angles, and discovering the rule for the angles inside a polygon.

---

1. The Parts of an Angle, and Its Name

An **angle** is made by two **arms** (rays) that start at the same point. That point is the **vertex**. The size of an angle is the amount of **turn** from one arm to the other. We measure it in **degrees** (°). One full turn is 360°, a half turn is 180° and a quarter turn is 90°.



Three ways to name an angle

1. Three letters, with the **vertex in the middle**: $\angle ABC$ or $\angle CBA$.

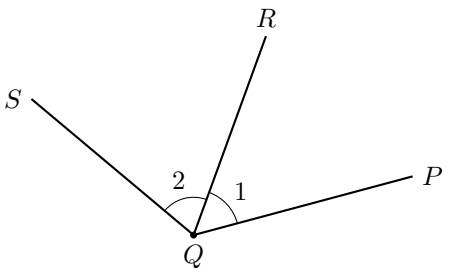
2. The vertex letter alone: $\angle B$. Use this only when there is just one angle at that vertex.

3. A small number or letter written inside the angle: $\angle 1$.

The symbol $\angle$ means "angle".

Watch out. The length of the arms does *not* change the size of an angle. Long arms or short arms, the turn between them is the same. If the arms are too short to measure, make them longer with a ruler.

Example 1



(a) Name $\angle 1$ in two other ways.

_____

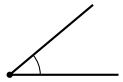
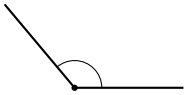
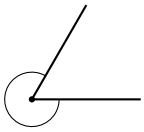
(b) Name the vertex and the two arms of $\angle 2$.

_____

(c) Why is it not clear to call $\angle 1$ "$\angle Q$"?

2. Types of Angles

We sort angles into types by their size. Every type has a name you need to know.

Acute	**Right**	**Obtuse**	**Straight**	**Reflex**
				
less than 90°	exactly 90°	between 90° and 180°	exactly 180°	between 180° and 360°
a sharp corner	a square corner (a small square marks 90°)	wider than a square corner	a straight line	more than a half turn

Every pair of arms really makes *two* angles: the small one inside, and the big one going the long way around. Together they make a full turn, 360°. When we say "the angle" we mean the smaller one, unless it is marked the long way around. Then it is a **reflex** angle.

Angles are all around us: the corner of a page, a door swinging open, a car's seat back, the slope of a roof. An angle can also be made where a **line meets a flat surface** (a **plane**), like a fence post standing in the ground or a ladder leaning on the floor. A fence post that is straight up meets the ground at 90°. A leaning ladder makes an acute angle with the floor.

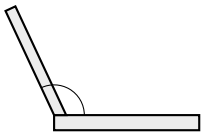
Example 2

Write the type of each angle: acute, right, obtuse, straight or reflex.

Angle	47°	90°	163°	181°	300°	89.5°	180°	112°
Type								

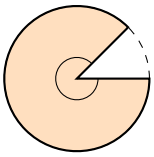
Example 3

Write the type of the marked angle in each picture.



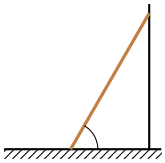
(a) A laptop screen and its keyboard.

_____



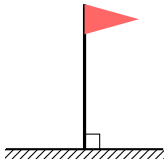
(b) The pizza that is left after one slice of 8 is eaten (the angle at the centre).

_____



(c) A ladder and the floor it stands on.

_____

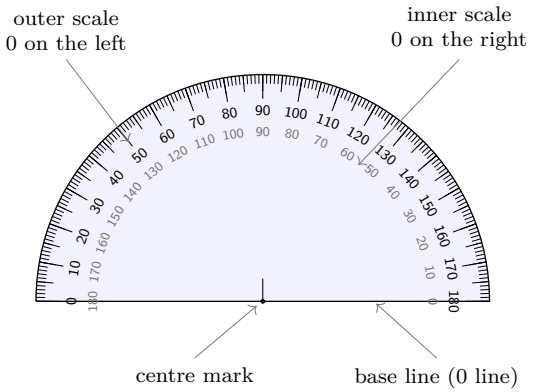


(d) A flagpole standing straight up and the ground.

_____

3. Measuring an Angle with a Protractor

A **protractor** measures angles. It has **two scales** of numbers that run in opposite directions. The **outer** (black) scale starts at 0 on the *left*. The **inner** (grey) scale starts at 0 on the *right*. The **centre mark** goes on the vertex, and the **base line** (the 0 line) goes along one arm.



Measuring an angle: 3 steps

Step 1. Put the **centre mark** exactly on the **vertex**.

Step 2. Turn the protractor so the **base line** lies along **one arm**.

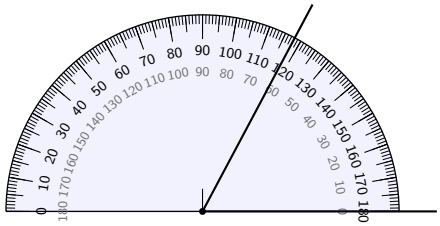
Step 3. Find the **0** that sits on that arm. Follow **that** scale around to the other arm and read the number.

Check: Does the angle look acute or obtuse? Every mark has two numbers (like 50 and 130). Pick the one that matches.

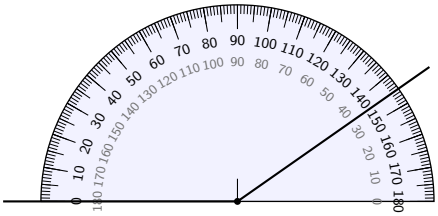
Watch out. The angle does not have to point to the right, and the protractor does not have to be level. Turn the protractor (or the page) so the base line lies on an arm. The 0 you start from can be on the left or on the right.

Example 4

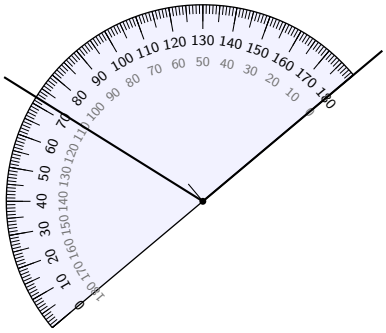
Read each protractor. Give the size of the angle and its type.



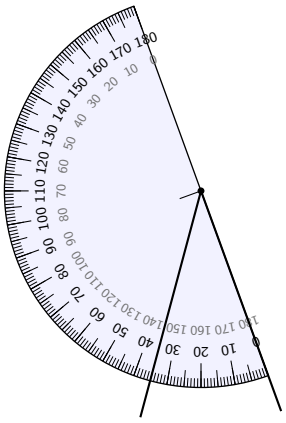
(a) Size _____ Type _____



(b) Size _____ Type _____



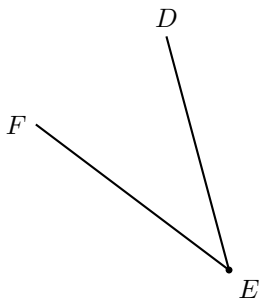
(c) Size _____ Type _____



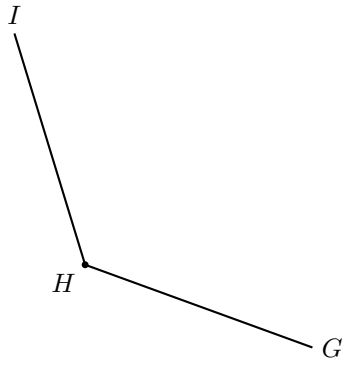
(d) Size _____ Type _____

Example 5

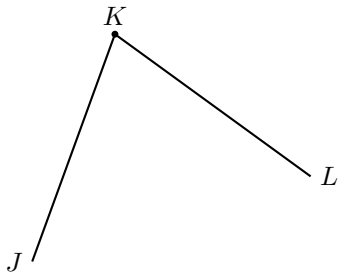
Use your own protractor to measure each angle. (The pictures are full size.) Then write its type.



$\angle DEF =$ _____ _____



$\angle GHI =$ _____ _____



$\angle JKL =$ _____ _____

4. Drawing Angles, and Reflex Angles

Drawing an angle of a given size (for example, 70°)
Step 1. Use a ruler to draw one arm. Put a dot at one end for the vertex.
Step 2. Put the centre mark on the vertex and the base line along the arm.
Step 3. Start at the 0 on your arm. Follow that scale to 70 and make a small dot at the edge.
Step 4. Take the protractor away. Join the vertex to the dot with a ruler. Label the angle 70°.

Example 6

Draw each angle on the arm given. Mark and label the angle.

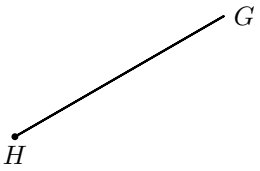
(a) $\angle ABC = 50°$



(b) $\angle DEF = 135°$



(c) $\angle GHI = 90°$



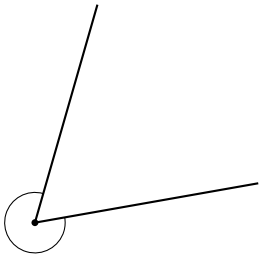
A protractor only goes up to 180°. To find a **reflex** angle, measure the *small* angle that goes with it, then subtract from 360°.

Reflex angle = 360° − the smaller angle Check: the answer must be between 180° and 360°.
To **draw** a reflex angle, first find the smaller angle (360° − reflex), draw it, and then mark the reflex angle with an arc going the long way around.

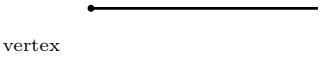
Example 7

(a) The smaller angle between two arms is 125°. What is the reflex angle?

(b) Measure the marked reflex angle.

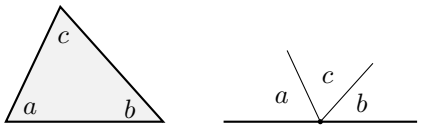


(c) Draw a reflex angle of 250° on the arm given.



5. Angles Inside a Polygon

A **polygon** is a closed shape made of straight sides. It is named by its number of sides, n: triangle (3), quadrilateral (4), pentagon (5), hexagon (6), heptagon (7), octagon (8), nonagon (9), decagon (10), dodecagon (12). The angles inside a polygon, at its corners, are its **interior angles**. A **regular** polygon has all sides equal and all angles equal.

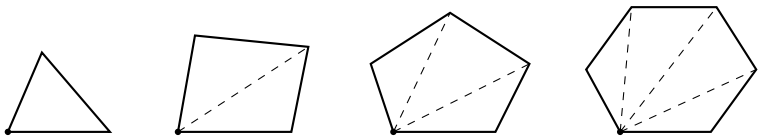


Triangles first. Cut out any paper triangle and tear off its three corners. Put the corners side by side: they always make a straight line. So the angles in *any* triangle add up to 180°.

Inductive reasoning means looking at several examples, finding a pattern, and using the pattern to make a general rule. We use it now to find a rule for any polygon.

Example 8

In each polygon, lines are drawn from *one* corner to the other corners. This splits the polygon into triangles.



(a) Complete the table. For 7 and 10 sides, use the pattern instead of drawing.

Number of sides, n	3	4	5	6	7	10	n
Number of triangles							
Sum of the angles							

(b) Describe the pattern in words.

Sum of the interior angles of a polygon with n sides:	$S = (n-2) \times 180°$
Each angle of a regular polygon:	$S \div n$

Calculator. Use brackets so the subtracting is done first. For 10 sides: (10 − 2) × 180 ENTER gives 1440.

Example 9

A stop sign is a regular octagon.



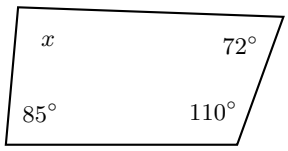
(a) Find the sum of its interior angles.

(b) A sign maker cuts the corners of a stop sign from sheet metal. What angle is each corner?

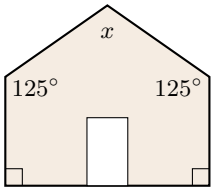
(c) What type of angle is each corner? _____

Example 10

Find the missing angle x in each shape.



(a) A quadrilateral.



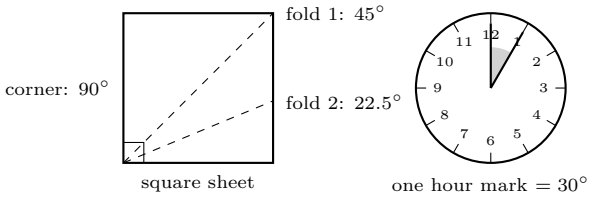
(b) The end wall of a garden shed is a pentagon. Find the angle at the peak of the roof.

Lesson 4: Estimating Angles with Referents, and True Bearings

Using a few angles you can picture in your head to estimate and sketch any angle, then using angles measured from north to describe directions on a map.

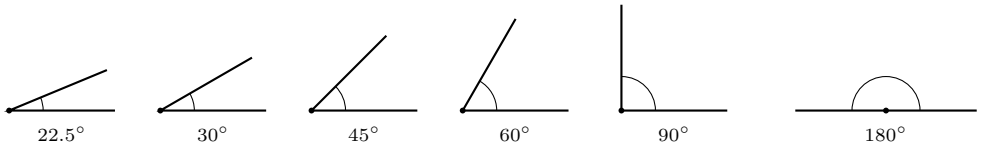
1. Referent Angles

A **referent angle** is an angle you can picture easily because you know something that has that size. You can make every referent with a sheet of paper or a clock.



Referent angles	
90°	corner of a sheet of paper
45°	fold the corner in half
22.5°	fold it in half again
30°	one hour mark on a clock
60°	two hour marks on a clock
180°	a straight line

A clock has 12 hour marks around a full turn of 360°, so each hour mark is $360° \div 12 = 30°$. Bigger angles are **built** from referents: $135° = 90° + 45°$, and $330° = 360° - 30°$.



Example 1

(a) Find the smaller angle between the hands of a clock at each time.

Time	Work	Angle
1:00		
2:00		
3:00		
4:00		
6:00		
8:00		

(b) Build each angle from referents. Then name its type.

Angle	Built from	Type
135°		
67.5°		
150°		
270°		
15°		
112.5°		

2. Estimating the Size of an Angle

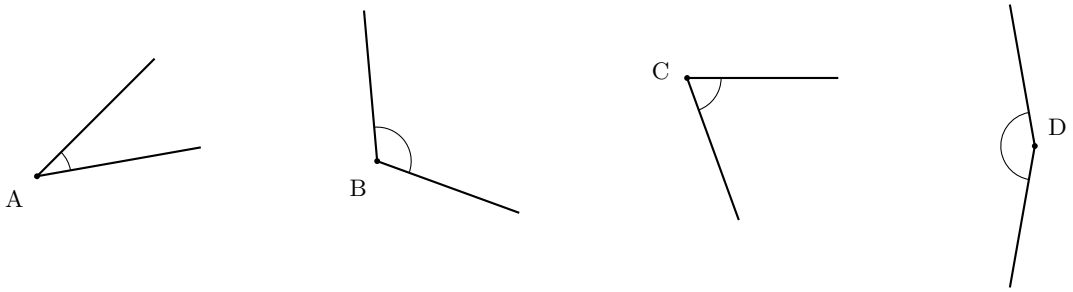
To **estimate** means to make a good guess using what you know. Estimate an angle *before* you measure it. Then you can tell right away if your protractor reading is wrong.

Estimating an angle
Step 1: Decide the type: less than 90° (acute), between 90° and 180° (obtuse), or more than 180° (reflex).
Step 2: Find the two referents it is between, and decide which one it is closer to.
Step 3: Write an estimate, such as "a little more than 60°, about 70°".

A good estimate is usually within about 10° of the measured angle. Turning the page does not change the size of an angle, so tilt your head or the page if it helps.

Example 2

Estimate each angle. Then measure it with a protractor.



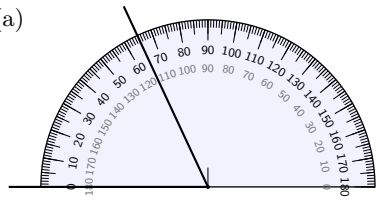
Angle	Type	Between which referents? Closer to?	Estimate	Measure
A				
B				
C				
D				

A protractor has two scales. Your estimate tells you which number to read.

Example 3

Estimate each angle first. Then read the protractor. The arm crosses two numbers: which one is the angle?

(a)

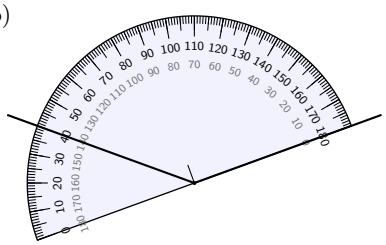


Estimate: _____

The two numbers: _____ and _____

The angle is _____.

(b)



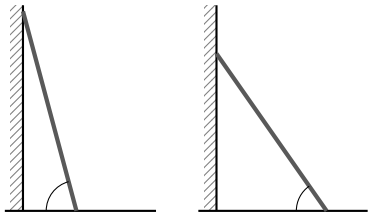
Estimate: _____

The two numbers: _____ and _____

The angle is _____.

Example 4

A safety rule for an extension ladder is that it should make an angle of about 75° with the ground. Too steep and it can tip back; too flat and the bottom can slide out.



(a) Estimate the angle between each ladder and the ground.
(b) Which ladder is set up safely? Explain.

3. Sketching an Angle

A **sketch** is a quick drawing without a protractor. Sketch an angle by building it from referents.

Sketching an angle
Step 1: Draw one arm and mark the vertex.
Step 2: Picture the referents nearby (90° is an L, 45° is half of that, 180° is a straight line).
Step 3: Draw the second arm where you think the angle is. Mark the angle with an arc.
Step 4: Check with a protractor. Were you within about 10°?

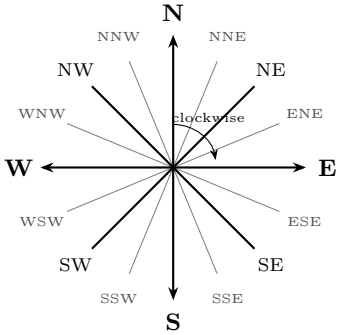
Example 5

Sketch each angle without a protractor. Write the referents you used. Then check with a protractor.

(a) 45° (b) 150° (c) 20° (d) 240°

4. True Bearings

A **true bearing** gives a direction as an angle. It is measured **clockwise from north**. Pilots, ship captains, hunters, surveyors and orienteers use bearings because a number is more exact than "northeast".



True bearing: the angle measured **clockwise from north**, written with **3 digits**.

N = 000°	E = 090°	S = 180°	W = 270°
NE = 045°	SE = 135°	SW = 225°	NW = 315°

Bearings go from 000° up to (but not including) 360°.

The 16-point compass splits each 45° in half again, so the points are 22.5° apart. NNE (north-northeast) is halfway between N and NE: 022.5°.

Watch out. Always start at north and turn **clockwise** (the way clock hands move). Write 3 digits: east is 090°, not 90°.

A direction is often described as "20° south of east". That means: face east, then turn 20° toward south. Make a quick sketch, start from the bearing of the main direction (east is 090°), and add if the turn is clockwise, subtract if it is

counterclockwise.

Example 6

(a) Write the true bearing of each compass direction.

Dir.	Bearing	Dir.	Bearing
E		NE	
S		SW	
W		NNE	
NW		WSW	

(b) Write each direction as a true bearing.

(i) 30° east of north _____

(ii) 20° south of east _____

(iii) 40° west of south _____

(iv) 15° west of north _____

5. Bearings on a Map, and the Bearing Back

A school protractor only goes to 180°. For a bearing more than 180°, measure the angle on the *west* side of the north line and subtract it from 360°.

Drawing or measuring a bearing from a point
Step 1: Draw a north line through the point (straight up, the same way as the map's north arrow).
Step 2: Put the protractor's centre on the point and its 0 line on the north line.
Step 3: If the path goes to the **east** (right) side: bearing = the angle you measure.
If it goes to the **west** (left) side: bearing = 360° − the angle you measure.
(To draw 250°: 360° − 250° = 110°, so mark 110° on the west side of north.)

Example 7

Draw a line from each point P in the direction of the bearing.

(a) 

060°

(b) 

135°

(c) 

250°

(d) 

320°

Example 8

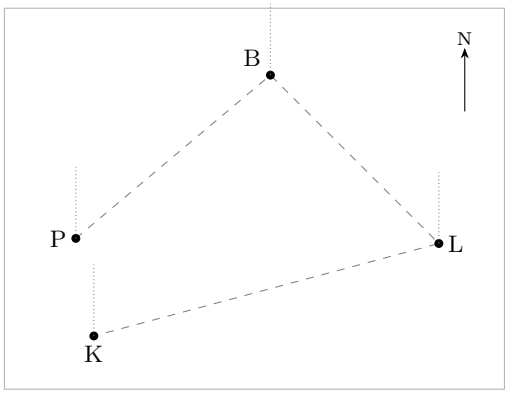
A hiking map in Kananaskis shows a parking lot (P), a bridge (B), a lookout (L) and a lake (K). Use a protractor.

(a) Find the bearing from P to B. _____

(b) Find the bearing from B to L. _____

(c) Find the bearing from L to K. _____

(d) The hikers walk from the lake straight back to the parking lot. What bearing do they follow? _____



The **bearing back** (also called the back bearing) is the direction for the return trip. It points exactly the opposite way, so it differs by 180°.

Bearing back: bearing less than 180°: **add** 180° bearing 180° or more: **subtract** 180°
Check: the answer must be between 000° and 359°.

Example 9

(a) Find the bearing back for each bearing.

Bearing	070°	145°	200°	315°	180°
Back					

(b) A bush pilot flies from Fort McMurray to a fishing lodge on a bearing of 335°. On what bearing should she fly to get home?

Lesson 5: Copying and Bisecting Angles; Perpendicular Lines

Copying an angle with a protractor, a compass or a carpenter's tool, cutting an angle exactly in half, drawing perpendicular lines, and sorting lines as parallel, perpendicular or neither.

1. Copying an Angle

To **copy** (or **replicate**) an angle means to draw a new angle with exactly the same size. Two angles with the same size are called **congruent**. Carpenters copy angles all the time: the angle of a stair, a roof or a crooked wall has to be copied onto a board before it is cut.

Tool	**How it copies an angle**
Protractor	Measure the angle. Then draw a new angle with the same number of degrees.
Tracing paper or a Mira	Trace the angle and move the tracing. A **Mira** is a see-through red plastic mirror: you look through it, line up the reflection of the angle, and draw it.
Sliding T-bevel	A carpenter's tool with a blade that swings and locks. Set the blade along both arms of the angle, lock it, then move it to the new board and draw.
Carpenter's square	An L-shaped steel square. It only copies a **right angle** (90°). A **speed square** (a triangle) also marks 45° and has a degree scale.
Compass and straightedge	No numbers at all: the compass copies distances, and that copies the angle (steps below).
Geometry software	Many computer drawing programs can measure an angle and copy it exactly.

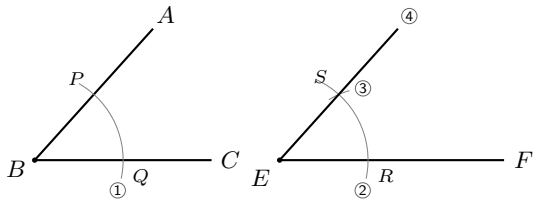
Copy $\angle ABC$ with a compass and straightedge

Step 1. Put the compass point on the vertex B. Draw an arc that crosses both arms. Label the crossings P and Q.

Step 2. Draw a ray from a new point E. *Keep the same compass width.* Put the point on E and draw a long arc that crosses the ray at R.

Step 3. Set the compass to the distance from Q to P. Put the point on R and draw an arc that crosses the long arc at S.

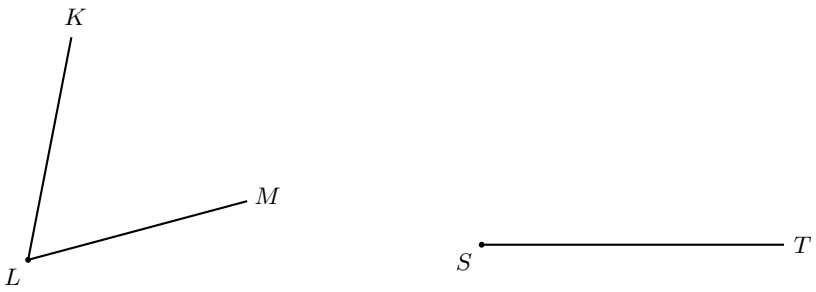
Step 4. Draw the ray from E through S. Now $\angle SER = \angle ABC$.



Watch out. In Steps 1 and 2 the compass width must be the **same**. Only in Step 3 do you change it, to the distance from Q to P. Hold the compass by the top so the width doesn't slip.

Example 1

Measure $\angle KLM$ with a protractor. Then use the protractor to copy it onto ray ST, with its vertex at S.



Example 2

Copy $\angle DEF$ onto ray GH using only a compass and a straightedge. Then measure both angles to check.



2. Bisecting an Angle

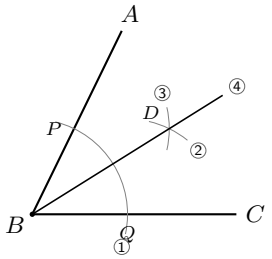
To **bisect** means to cut into two equal parts. The **angle bisector** is the ray from the vertex that cuts an angle into two equal angles. Each half is the angle $\div 2$. If you know one half, the whole angle is the half $\times 2$.

Ways to bisect an angle:

- **Protractor:** measure, divide by 2, and mark that many degrees from one arm.
- **Folding:** draw or trace the angle on thin paper. Fold through the vertex so one arm lies exactly on top of the other. The crease is the bisector.
- **Mira:** stand it on the vertex and turn it until the reflection of one arm lies on the other arm. Draw along its edge.
- **Compass and straightedge:** the steps below.

Bisect $\angle ABC$ with a compass and straightedge

Step 1. Compass point on the vertex B. Draw an arc that crosses both arms at P and Q.
Step 2. Compass point on Q. Draw an arc inside the angle.
Step 3. *Same width.* Compass point on P. Draw an arc that crosses the Step 2 arc at D.
Step 4. Draw the ray from B through D. Ray BD bisects $\angle ABC$, so $\angle ABD = \angle DBC$.



Watch out. In Steps 2 and 3, the compass must open to **more than half** the distance from P to Q, or the two arcs won't cross. Don't change the width between Step 2 and Step 3.

Example 3

Each angle is bisected. Complete the table.

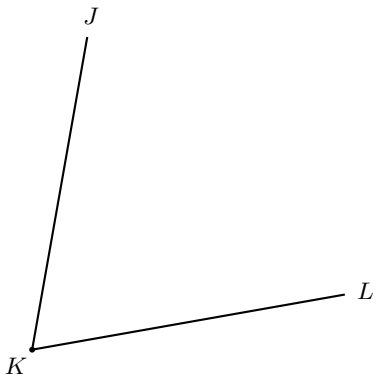
	(a)	(b)	(c)	(d)	(e)	(f)
Whole angle	90°	128°	75°	180°		
Each half					34°	81.5°

Example 4

(a) Bisect $\angle JKL$ with a compass and straightedge.

(b) Measure the two halves with a protractor. What should each one be?

(c) How could you check your bisector by folding?

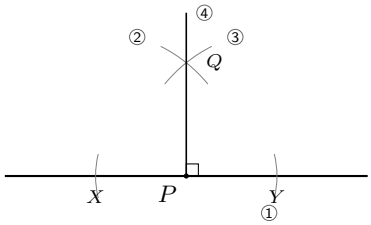


3. Perpendicular Lines and the Perpendicular Bisector

Two lines are **perpendicular** if they meet at a right angle (90°). We mark the right angle with a small square, and write $AB \perp CD$ ("AB is perpendicular to CD"). When two lines are perpendicular, all four angles where they cross are 90°.

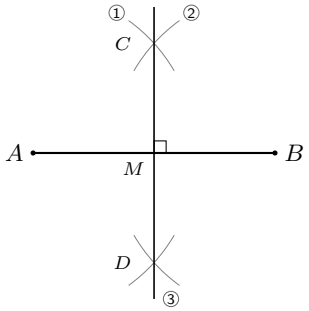
A straight angle is 180°, and $180° \div 2 = 90°$. So drawing a perpendicular line is the same as **bisecting a straight angle**. The **midpoint** of a segment cuts it into two equal parts. The **perpendicular bisector** of a segment goes through its midpoint at 90°.

Perpendicular through a point P on a line



1. Compass point on P. Draw arcs that cross the line on both sides, at X and Y.
2. Open the compass wider. Point on X: draw an arc above P.
3. Same width, point on Y: draw an arc that crosses it at Q.
4. Draw line PQ. It is perpendicular to the first line.

Perpendicular bisector of segment AB



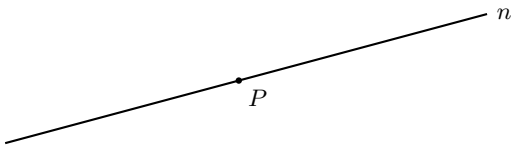
1. Open the compass to *more than half* of AB. Point on A: draw arcs above and below AB.
2. Same width, point on B: draw arcs that cross the first two at C and D.
3. Draw line CD. It crosses AB at its midpoint M, at 90°. So $AM = MB$.

On the job. A carpenter's (framing) square has a perfect 90° corner. Hold one edge along a board and draw along the other edge: the line is perpendicular to the board's edge. You can also fold a paper line onto itself: the crease is perpendicular to the line.

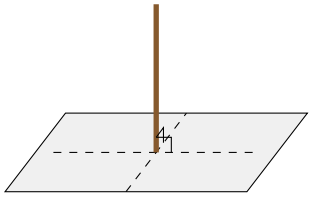
Example 5

(a) Construct the perpendicular bisector of segment AB. Label the midpoint M. Measure AM and MB.

(b) Construct a line through P that is perpendicular to line n.



A post, a pole or a table leg meets a flat surface (a **plane**), not just a line. It is perpendicular to the surface only if it makes 90° with *every* line on the surface through its foot. In practice, a builder checks with a square or a level in **two directions**: front-to-back and side-to-side.



The post is perpendicular to the ground:
90° to the side-to-side line *and*
90° to the front-to-back line.

Example 6

Mateo is building a table. He checks one leg and finds that it makes an angle of 92.5° with the underside of the tabletop.

(a) Is the leg perpendicular to the tabletop? Explain.

(b) By how many degrees must he move the leg?

(c) The leg and the tabletop make two angles side by side, on a straight line. What is the other angle?

(d) The leg is now 90° from front to back. Why should Mateo also check it from side to side?

4. Parallel, Perpendicular or Neither?

Parallel lines lie on the same flat surface and **never meet**. They stay the same distance apart, like the two rails of a railway track. We mark parallel lines with matching arrows (>) and write $AB \parallel CD$.

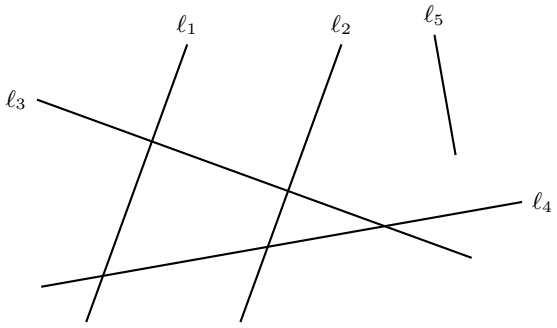
Sorting two lines

1. Do the lines meet, or would they meet if you made them longer? If **no**, they are **parallel**. (Check: the distance between them, measured straight across, is the same in two places.)
2. If **yes**, measure the angle where they meet. 90° → **perpendicular**. Any other angle → **neither**.

Watch out. A line goes on forever. Two lines that don't touch *on the page* may still meet if you extend them with a ruler. Then they are not parallel.

Example 7

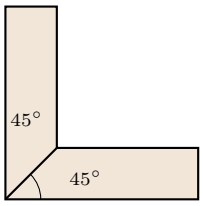
Use a ruler and protractor. Sort each pair of lines as parallel, perpendicular or neither, and give a reason.



Pair	Sort	Reason
ℓ_1, ℓ_2		
ℓ_1, ℓ_3		
ℓ_2, ℓ_3		
ℓ_1, ℓ_4		
ℓ_3, ℓ_4		
ℓ_4, ℓ_5		

5. On the Job: Mitre Cuts and Braces

A **mitre joint** joins two boards at a corner, like the corner of a picture frame. Each board end is cut on an angle. For a tight corner, the joint line must **bisect** the corner angle, so **each cut is half the corner angle**.



Mitre cut angle = corner angle $\div 2$

Square or rectangular frame: $90° \div 2 = 45°$.

For a frame shaped like a regular polygon with n sides, find one corner first (Lesson 3): corner $= (n-2) \times 180° \div n$.

On the job. A mitre saw's scale shows how far the blade is turned away from a square (90°) cut. So the **saw setting** $= 90° -$ the cut angle. For a 45° cut, the setting is also 45°.

A **brace** is a board nailed across a corner to keep a gate or shelf from sagging. On a square gate, a brace from corner to corner bisects the 90° corners.

Example 8

(a) Ava is making a rectangular picture frame. At what angle does she cut the end of each piece?

(b) Jonah is making a frame for a mirror shaped like a regular hexagon (6 sides). Find the corner angle, the cut angle, and the mitre saw setting.

(c) A square garden gate has a brace from the bottom hinge corner to the top latch corner. What angle does the brace make with the top rail? Why?

Lesson 6: Angle Partners: Complementary, Supplementary and Opposite Angles

Angles that sit next to each other, angles that add to 90° or 180°, the equal angles made by two crossing lines, and finding missing angles one step at a time with a reason for each step.

1. Adjacent Angles

An angle has two **arms** that meet at a point called the **vertex**. You can name an angle three ways: with three letters and the vertex in the middle ($\angle PQR$), with the vertex letter alone ($\angle Q$), or with a small number or letter written inside the corner ($\angle 1$ or $\angle a$).

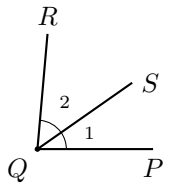
Two angles are **adjacent** ("next to each other") when they

- have the **same vertex**,
- share **one arm**, and
- do **not overlap** (one angle is not inside the other).

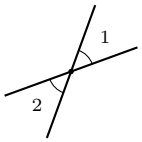
Adjacent angles add up. If $\angle PQS$ and $\angle SQR$ are adjacent, then $\angle PQS + \angle SQR = \angle PQR$ (the big angle made by the two outside arms).

Example 1

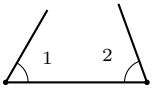
In each picture, are $\angle 1$ and $\angle 2$ adjacent? Write yes or no, and give a reason.



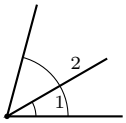
(a) Adjacent? _____



(b) Adjacent? _____



(c) Adjacent? _____

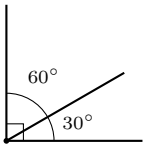


(d) Adjacent? _____

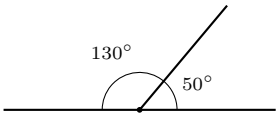
(e) In picture (a), $\angle PQS = 35°$ and $\angle SQR = 50°$. Find $\angle PQR$.

2. Complementary and Supplementary Angles

Two angles are **complementary** if they add up to 90°. Each angle is the **complement** of the other. Two angles are **supplementary** if they add up to 180°. Each angle is the **supplement** of the other.



complementary: $30° + 60° = 90°$



supplementary: $50° + 130° = 180°$

Complement = 90° − the angle **Supplement** = 180° − the angle

Only an **acute** angle (less than 90°) has a complement. Only an angle less than 180° has a supplement.

Remember. Complementary goes with a **C**orner (90°). **S**upplementary goes with a **S**traight line (180°). The two angles do **not** have to be next to each other: a 30° angle in one corner of the page and a 60° angle in another corner are still complementary.

Example 2

Find the complement and the supplement of each angle. If one does not exist, say why.

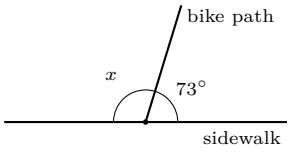
Angle	**Complement** (90° − angle)	**Supplement** (180° − angle)
(a) 25°		
(b) 64°		
(c) 8°		
(d) 97°		
(e) 210°		

Example 3

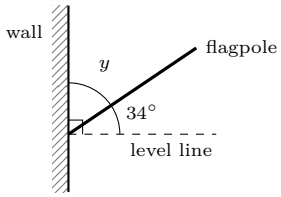
(a) The complement of an angle is 31°. What is the angle? What is its supplement?

(b) The supplement of an angle is 142°. What is the angle? What is its complement?

(c) A straight bike path ends at a straight sidewalk. The angle on one side of the path is 73°. What is the angle x on the other side?



(d) A flagpole holder is screwed to a straight-up (vertical) wall. The pole makes a 34° angle with a level line out from the wall. What angle y does the pole make with the wall above it?



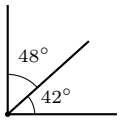
3. Complementary, Supplementary or Neither?

Adjacent angles are **not always** complementary or supplementary. Add them to find out:

Sum is 90° → complementary (the outside arms make a square corner).
Sum is 180° → supplementary (the outside arms make a straight line).
Any other sum → neither.

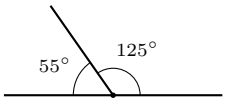
Example 4

Each picture shows two adjacent angles. Find the sum. Then write *complementary*, *supplementary* or *neither*.

(a) 

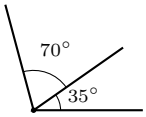
Sum: _____

_____

(b) 

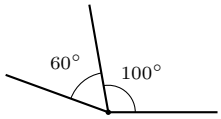
Sum: _____

_____

(c) 

Sum: _____

_____

(d) 

Sum: _____

_____

(e) Can two angles be supplementary if they are **not** adjacent? Give an example.

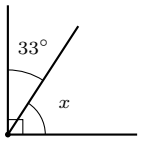
To find a **missing angle** in a corner or on a straight line:

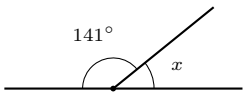
Step 1: Decide: is it a right angle (90°), a straight line (180°), or neither?
Step 2: Write an equation: (missing angle) + (the angles you know) = 90° or 180°.
Step 3: Subtract to find the missing angle. **Step 4:** Check by adding.

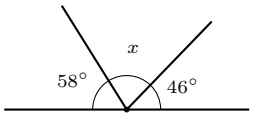
Watch out. Don't guess from how a picture looks. A corner is 90° only if it has the small square mark (or the question says so).

Example 5

Find each missing angle x. Write the equation first.

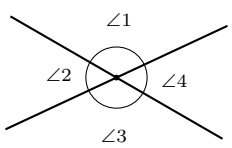
(a) 

(b) 

(c) 

4. Vertically Opposite Angles and Angles Around a Point

When two straight lines cross, they make four angles. The two angles **across** from each other are called **vertically opposite angles**. ("Vertically" means they share a *vertex*. It does not mean up and down.) Vertically opposite angles are always **equal**.

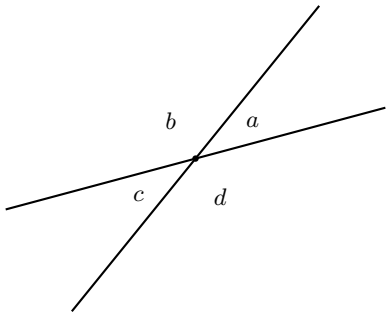


Why are they equal?
$\angle 1 + \angle 2 = 180°$ (they make a straight line)
$\angle 3 + \angle 2 = 180°$ (they also make a straight line)
So $\angle 1$ and $\angle 3$ are both "180° minus $\angle 2$". That means $\angle 1 = \angle 3$. In the same way, $\angle 2 = \angle 4$.

A full turn is 360°, so all the angles **around a point** add up to 360°. Each angle needs to be next to the one before it, with no gaps and no overlaps.

The four angle facts (use these words as your reasons)	
Complementary angles	add up to 90° (a right angle)
Supplementary angles (angles on a straight line)	add up to 180°
Vertically opposite angles	are equal
Angles around a point	add up to 360°

Example 6



(a) Measure each angle with a protractor.

$\angle a =$ _____ $\angle b =$ _____ $\angle c =$ _____ $\angle d =$ _____

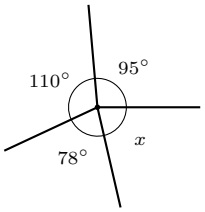
(b) Which angles are equal? Why?

(c) Add all four. What do you get? Why?

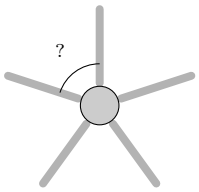
(d) Suppose you only knew $\angle a = 36°$. How could you find the other three without a protractor?

Example 7

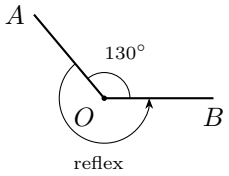
(a) Find x.



(b) A ceiling fan has 5 blades, evenly spaced. What angle is between two blades that are next to each other?



(c) $\angle AOB = 130°$. The **reflex** angle AOB goes the long way around. How big is it?



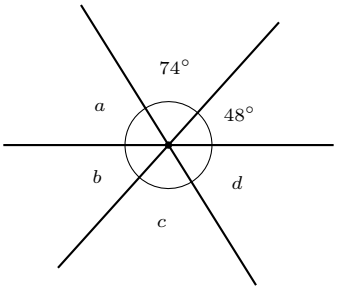
5. Finding Missing Angles with Reasons

In a bigger diagram, you find the angles **one at a time**. Each new angle you find helps you find the next one.

Step 1: Look for a missing angle that is next to or across from an angle you know.
Step 2: Find it, and write the **reason** (one of the four angle facts).
Step 3: Repeat with the new angle. **Check:** the angles around the point add up to 360°.

Example 8

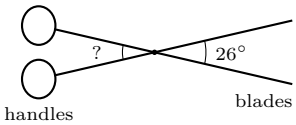
Three straight lines cross at one point. Find angles a, b, c and d. Give a reason for each.



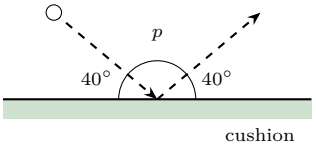
Angle	Work	Reason
a		
b		
c		
d		

Example 9

(a) A pair of scissors is open. The angle between the blades is 26°. What is the angle between the handles? Why?



(b) In a pool game, a ball hits the straight cushion (edge) of the table and bounces off. It comes in at 40° to the cushion and leaves at 40° to the cushion. Find the angle p between its two paths.



Lesson 7: Transversals: Naming the Angle Pairs

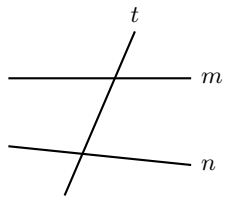
When one line crosses two other lines it makes eight angles. Learning the names of the angle pairs, finding them in diagrams and in real pictures, and seeing why the lines must be parallel before the pairs are equal.

1. Transversals and the Eight Angles

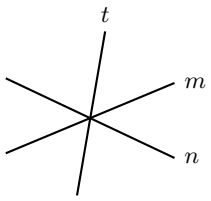
A **transversal** is a line that crosses two (or more) other lines at **two different points**. In the diagrams, line t is the transversal and ℓ_1 and ℓ_2 (say "line one" and "line two") are the two lines it crosses. Remember that a line goes on forever; we only draw part of it.

Example 1

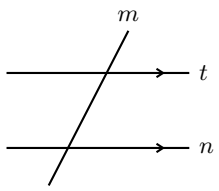
Is line t a transversal of lines m and n? Write yes or no, and give a reason.



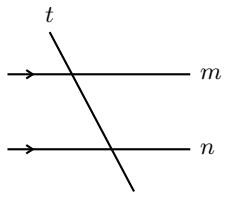
(a) _____



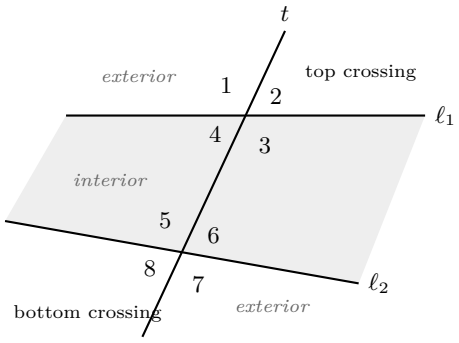
(b) _____



(c) _____



(d) _____



A transversal crossing two lines makes **eight angles**, four at each crossing.

How we number the eight angles (every diagram in this unit):
At the **top crossing**, angles 1, 2, 3, 4 go **clockwise**, starting at the **upper left**.
At the **bottom crossing**, angles 5, 6, 7, 8 go **clockwise**, starting at the **upper left**.

Interior angles are *between* the two lines (the shaded band): $\angle 3$, $\angle 4$, $\angle 5$ and $\angle 6$.
Exterior angles are *outside* the two lines: $\angle 1$, $\angle 2$, $\angle 7$ and $\angle 8$.
Each angle is also on the **left** or the **right side** of the transversal.

Good to know. The two lines do not have to be parallel. Here ℓ_1 and ℓ_2 are not parallel, but the names in this lesson still work. When lines are parallel, we mark them with matching arrows (>), as in (c) and (d) above.

Example 2

Use the diagram above. Complete the table for all eight angles.

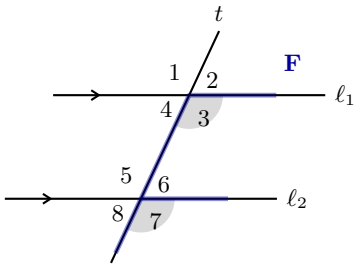
Angle	$\angle 1$	$\angle 2$	$\angle 3$	$\angle 4$	$\angle 5$	$\angle 6$	$\angle 7$	$\angle 8$
Top or bottom crossing?								
Interior or exterior?								
Left or right of t?								

2. Corresponding and Vertically Opposite Angles

Corresponding angles are in the **same position** at the two crossings: both upper left, both upper right, both lower right or both lower left. They are on the same side of the transversal, and one is interior while the other is exterior.

Corresponding angles make an **F shape**. The F can be backwards, upside down or tilted.

In our numbering, the four pairs are easy to remember: add 4.
$\angle 1$ and $\angle 5$, $\angle 2$ and $\angle 6$, $\angle 3$ and $\angle 7$, $\angle 4$ and $\angle 8$.

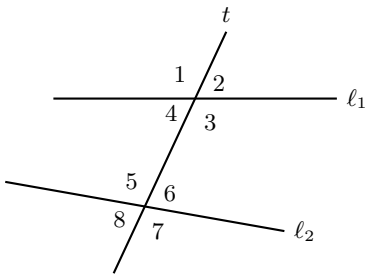


$\angle 3$ and $\angle 7$ are corresponding.

Vertically opposite angles (Lesson 6) are **across from each other at the same crossing**, like the opposite parts of an X. They are **always equal**. The pairs are $\angle 1$ and $\angle 3$, $\angle 2$ and $\angle 4$, $\angle 5$ and $\angle 7$, $\angle 6$ and $\angle 8$.

Watch out. Corresponding angles are at **different** crossings. Vertically opposite angles are at the **same** crossing. Two angles that sit next to each other at one crossing (like $\angle 1$ and $\angle 2$) make a straight line, so they are supplementary (they add to 180°).

Example 3



(a) The angle corresponding to $\angle 2$ is _____.

(b) The angle corresponding to $\angle 8$ is _____.

(c) The angle vertically opposite $\angle 5$ is _____.

(d) The angle vertically opposite $\angle 2$ is _____.

(e) $\angle 6$ and $\angle 8$ are _____.

(f) $\angle 4$ and $\angle 1$ are next to each other on a straight line. Together they make _____.

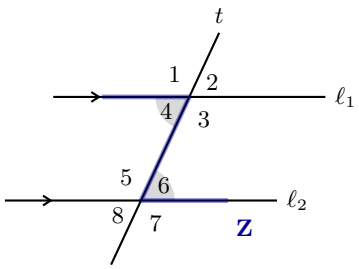
3. Alternate Angles and Same-Side Angles

Alternate means "one on each side". Alternate angles are at **different crossings** and on **opposite sides** of the transversal.

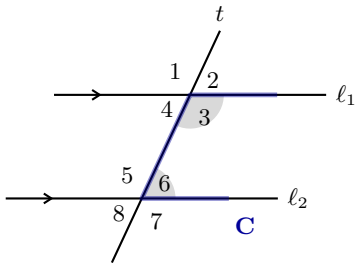
- **Alternate interior angles** are both *between* the lines, on opposite sides of t. They make a **Z shape**. Pairs: $\angle 3$ and $\angle 5$, $\angle 4$ and $\angle 6$.
- **Alternate exterior angles** are both *outside* the lines, on opposite sides of t. Pairs: $\angle 1$ and $\angle 7$, $\angle 2$ and $\angle 8$.

Two more pairs are at different crossings and on the **same side** of the transversal.

- **Interior angles on the same side of the transversal** are both between the lines, on the same side of t. They make a **C or U shape**. Pairs: $\angle 3$ and $\angle 6$, $\angle 4$ and $\angle 5$.
- **Exterior angles on the same side of the transversal** are both outside the lines, on the same side of t. Pairs: $\angle 1$ and $\angle 8$, $\angle 2$ and $\angle 7$.



alternate interior: $\angle 4$ and $\angle 6$



interior, same side: $\angle 3$ and $\angle 6$

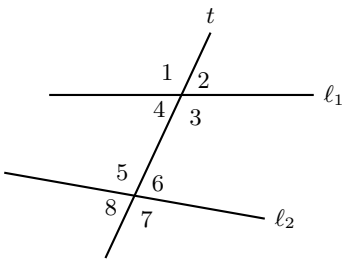
Naming a pair of angles: ask three questions.

1. Are the angles at the **same crossing**? If yes: across from each other → **vertically opposite**; next to each other → they make a **straight line** (180°).

If they are at **different crossings**, ask: **2.** Interior or exterior? **3.** Same side of t or opposite sides?

Interior or exterior?	**Side of t**	**Name**	**Pairs in our diagram**
one of each	same side	corresponding (F)	1 & 5, 2 & 6, 3 & 7, 4 & 8
both interior	opposite sides	alternate interior (Z)	3 & 5, 4 & 6
both exterior	opposite sides	alternate exterior	1 & 7, 2 & 8
both interior	same side	interior angles on the same side (C or U)	3 & 6, 4 & 5
both exterior	same side	exterior angles on the same side	1 & 8, 2 & 7
one of each	opposite sides	no special name	1 & 6, 2 & 5, 3 & 8, 4 & 7

Example 4



(a) Use the three questions to name each pair.

Pair	Same or different crossings?	Interior or exterior?	Side of t	Name
3 & 5				
2 & 8				
3 & 6				
1 & 8				
1 & 5				
2 & 3				
1 & 6				

(b) Name the angle that is:

alternate interior to $\angle 4$: _____

exterior on the same side as $\angle 7$: _____

interior on the same side as $\angle 5$: _____

alternate exterior to $\angle 1$: _____

(c) Kai says, "$\angle 3$ and $\angle 4$ are alternate interior angles, because both are between the lines and they are on opposite sides of t." What is wrong with Kai's reasoning?

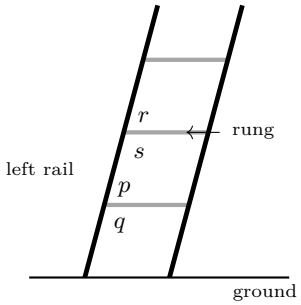
4. Real Pictures and Busy Diagrams

Real pictures are turned and tilted, and the angles may be numbered or lettered in any order. Look at **where** each angle is, not at its number.

How to find the transversal. The transversal is the line that goes through the **vertices of both angles**. The other line at each vertex is one of "the two lines". It can help to turn your page so the transversal goes up and down.

Example 5

(a) Part of a ladder. Its rungs are parallel.



(i) For $\angle p$, $\angle q$, $\angle r$ and $\angle s$, what are "the two lines" and what is the transversal?

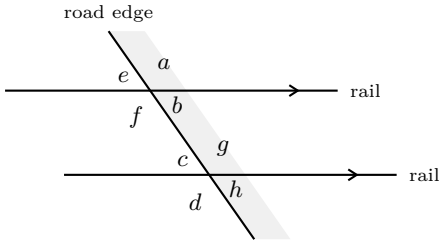
(ii) Name each pair.

$\angle p$ and $\angle r$: _____

$\angle p$ and $\angle s$: _____

$\angle q$ and $\angle r$: _____

(b) A road crosses two parallel railway rails.



Name each pair of angles.

(i) $\angle a$ and $\angle g$: _____

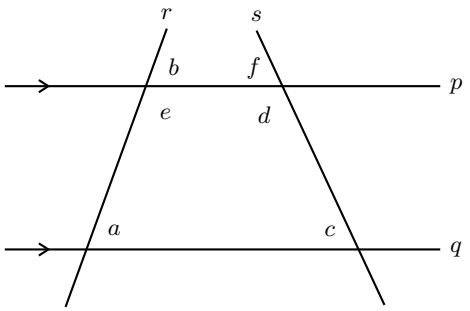
(ii) $\angle f$ and $\angle g$: _____

(iii) $\angle e$ and $\angle h$: _____

(iv) $\angle b$ and $\angle g$: _____

Example 6

A map shows four streets. 1st Avenue (p) and 2nd Avenue (q) are parallel. Centre Street (r) and River Road (s) cross both avenues.



For each pair, name the two lines and the transversal. Then name the pair.

Pair	Two lines	Transversal	Name
a & b			
c & d			
b & d			

(d) Using lines r and s with transversal p, which angle is alternate interior to $\angle e$? _____

5. When the Lines Are Not Parallel

The names tell you **where** two angles are. They do **not** tell you that the angles are the same size. Use a protractor to see what happens when the lines are parallel and when they are not.

Example 7

Measure the angles in each diagram with a protractor. (Extend the lines with a ruler if it helps.) Complete the table.

Diagram A

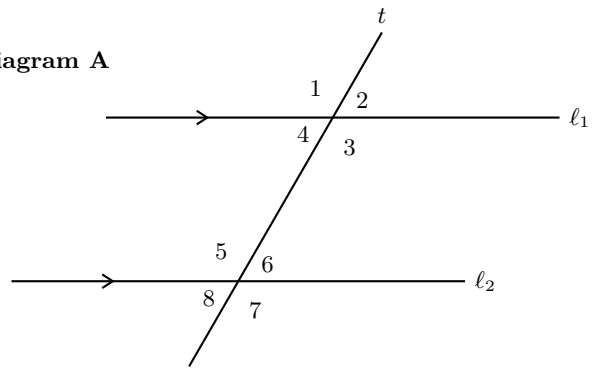
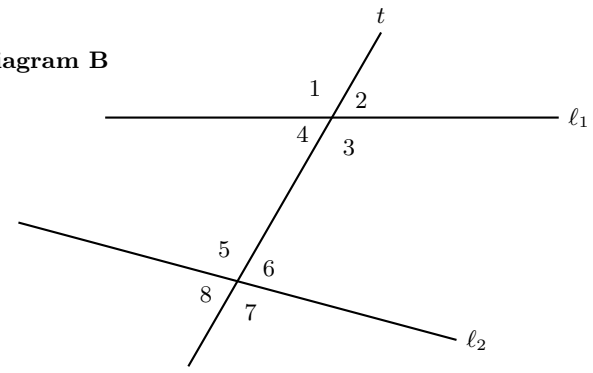


Diagram B



	corresponding			**alternate interior**		**interior, same side**
	$\angle 1$	$\angle 5$	Equal?	$\angle 4$	$\angle 6$	$\angle 4 + \angle 5$
A (parallel)						
B (not parallel)						

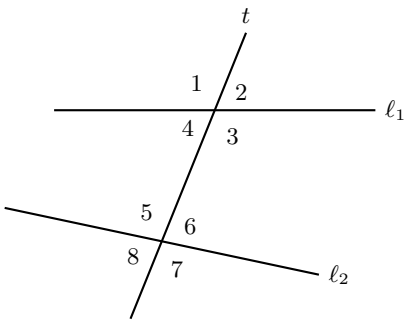
When the lines are parallel, corresponding angles are _____, alternate interior angles are _____, and interior angles on the same side add to _____. When the lines are _____, these pairs are not equal and the same-side pair does not add to 180°.

The names work for any two lines. The sizes depend on whether the lines are parallel.
Always true: vertically opposite angles are equal; two angles next to each other on a straight line add to 180°.
Only when the two lines are parallel (Lesson 8): corresponding angles, alternate interior angles and alternate exterior angles are equal; interior angles on the same side and exterior angles on the same side add to 180°.

Watch out. Never use the parallel-line rules unless you *know* the lines are parallel: the question says so, or the lines have arrow marks. Lines that only *look* parallel might not be.

Example 8

In this diagram, ℓ_1 and ℓ_2 are **not** parallel. $\angle 2 = 68°$ and $\angle 7 = 100°$.



(a) Find the other six angles. Give a reason for each one.

Angle	**Size**	**Reason**
$\angle 4$		
$\angle 1$		
$\angle 3$		
$\angle 5$		
$\angle 6$		
$\angle 8$		

(b) $\angle 2$ and $\angle 6$ are corresponding angles. Are they equal? What about the alternate interior angles $\angle 4$ and $\angle 6$?

(c) Why did we need one angle from *each* crossing to find all eight?

Lesson 8: Parallel Lines and Transversals: Finding Missing Angles

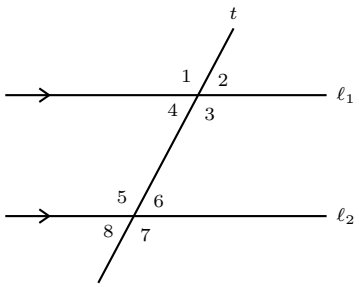
When the lines are parallel, the angle pairs follow rules: use them to find every missing angle, give a reason for each step, and check whether two lines really are parallel.

---

1. The Parallel-Line Rules

A **transversal** is a line that crosses two other lines. It makes 8 angles. As in Lesson 7, we number them like this: angles 1 to 4 are at the top crossing and angles 5 to 8 are at the bottom crossing. Each set is numbered **clockwise, starting at the upper-left**.

Two lines are **parallel** if they are always the same distance apart and never meet. On a diagram, matching arrow marks (>) on two lines mean they are parallel. We write $\ell_1 \parallel \ell_2$ (say "ℓ_1 is parallel to ℓ_2").



If $\ell_1 \parallel \ell_2$, these pairs are **equal**:

corresponding angles (F shape)	$\angle 1 = \angle 5,\ \angle 2 = \angle 6,\ \angle 3 = \angle 7,\ \angle 4 = \angle 8$
alternate interior angles (Z shape)	$\angle 3 = \angle 5,\ \angle 4 = \angle 6$
alternate exterior angles	$\angle 1 = \angle 7,\ \angle 2 = \angle 8$

and these pairs are **supplementary** (add to 180°):

interior angles on the same side (U shape)	$\angle 3 + \angle 6 = 180°,\ \angle 4 + \angle 5 = 180°$
exterior angles on the same side	$\angle 1 + \angle 8 = 180°,\ \angle 2 + \angle 7 = 180°$

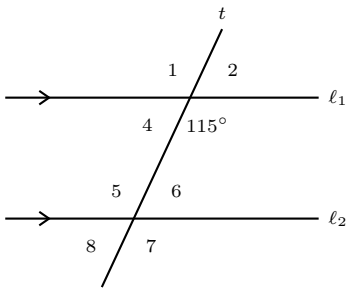
Always true, parallel or not: **vertically opposite** angles are equal ($\angle 1 = \angle 3$, $\angle 2 = \angle 4$, $\angle 5 = \angle 7$, $\angle 6 = \angle 8$), and two angles side by side on a **straight line** add to 180° (for example $\angle 1 + \angle 2 = 180°$).

Shortcut to check. With parallel lines there are only **two sizes** of angle. All the small (acute) angles are equal, all the big (obtuse) angles are equal, and small + big = 180°. Use this to check your answers, but still write the name of the rule as your reason.

Watch out. Use the parallel-line rules only when you *know* the lines are parallel: arrow marks, or the question says so. Lesson 7 showed that the pairs are not equal when the lines are not parallel.

Example 1

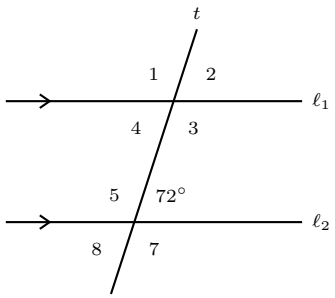
$\ell_1 \parallel \ell_2$ and $\angle 3 = 115°$. Complete the table.



Angle	How is it paired with $\angle 3$?	Equal, or add to $180°$?	Size
$\angle 5$			
$\angle 7$			
$\angle 1$			
$\angle 6$			
$\angle 4$			

Example 2

$\ell_1 \parallel \ell_2$ and $\angle 6 = 72°$. Find the other 7 angles. Give a reason for each one.



Angle	Size	Reason
$\angle 8$		
$\angle 2$		
$\angle 4$		
$\angle 5$		
$\angle 7$		
$\angle 1$		
$\angle 3$		

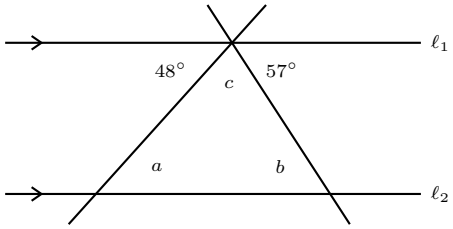
2. Finding Angles in Steps, With Reasons

Many diagrams need two or three steps. Look for an angle you *can* find, write its size on the diagram, then use it to find the next one. The **order** matters, so write your steps in a table: the angle, its size and the reason.

When an angle is named with a letter, such as x, write an equation and solve it. If x and $47°$ are on a straight line, then $x + 47° = 180°$, so $x = 180° - 47° = 133°$.

Example 3

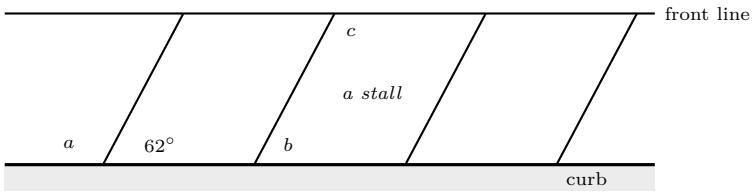
$\ell_1 \parallel \ell_2$. Two straight roads meet at point A on ℓ_1 and make a triangle with ℓ_2. Find a, b and c, in that order. Give a reason for each.



Angle	Size	Reason
a		
b		
c		

Example 4

In a parking lot, the stall lines are **parallel**. The curb is straight, and the painted line at the front of the stalls is parallel to the curb. Each stall line meets the curb at 62°.



(a) Write an equation for a. Solve it. (b) Find b. Give your reason. (c) Find c. Give your reason.

3. Perpendicular Transversals

Two lines are **perpendicular** if they meet at a right angle (90°). A small square in the corner shows a right angle. We write $t \perp \ell_1$.

If a transversal is perpendicular to one of two parallel lines, it is perpendicular to the other one too. The corresponding angles are equal, so they are both 90°. In fact, all 8 angles are 90°.

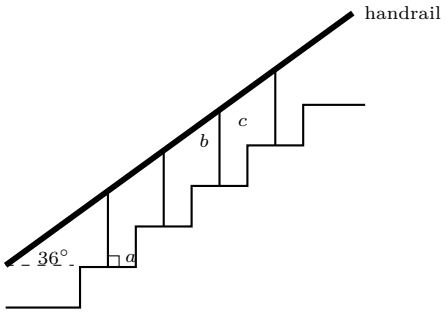
If $\ell_1 \parallel \ell_2$ and $t \perp \ell_1$, then $t \perp \ell_2$. All 8 angles are 90°. Two angles that make a right angle together are **complementary**: they add to 90°.

Builders use this. The wall studs in a house are vertical, so they are parallel to each other and perpendicular to the level floor.

Example 5

A carpenter is putting the balusters (the upright posts) on a staircase. The balusters are vertical, so they are parallel to each other and perpendicular to the level floor and stair treads. The handrail rises at 36° from the horizontal.

(a) What is angle a, where a baluster meets a tread? Why?



(b) Find b and c, the angles where a baluster meets the handrail.

(c) Why can the carpenter cut the tops of *all* the balusters at the same angle?

4. Are the Lines Parallel?

The rules also work **backwards**. Measure a pair of angles made by a transversal. If the pair follows a parallel-line rule, the lines are parallel. If it does not, the lines are **not** parallel: if you made them longer, they would meet.

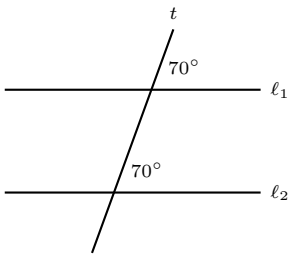
Two lines cut by a transversal **are parallel** if
- a pair of corresponding angles is equal, **or**
- a pair of alternate interior (or alternate exterior) angles is equal, **or**
- a pair of interior (or exterior) angles on the same side adds to 180°.

If the pair does not follow the rule, even by 1°, the lines are **not** parallel.

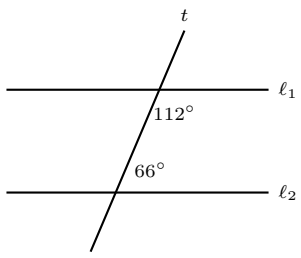
Watch out. Vertically opposite angles are *always* equal, so they tell you nothing about whether the lines are parallel. Also, don't trust your eyes: a diagram may not be drawn to scale. Use the angle sizes.

Example 6

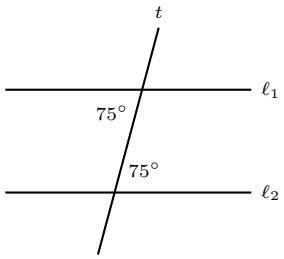
Leah is building a leaning bookshelf. The shelves are ℓ_1 and ℓ_2 and the slanted leg is the transversal t. She measures two angles. Are the shelves parallel? Give the rule you used. (The diagrams are not drawn to scale.)



(a)



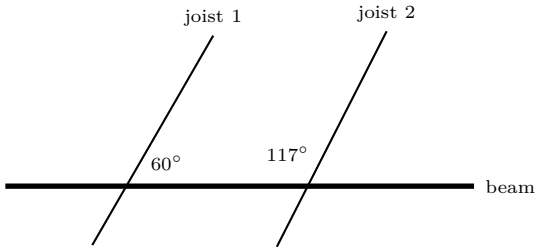
(b)



(c)

Example 7

Owen is building a deck at an angle. Two floor joists are nailed across a beam, and they must be parallel. He measures the angles shown.



(a) Are the joists parallel? Explain.

(b) What should the 117° angle be, so that the joists are parallel?

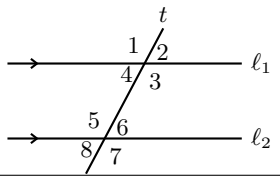
(c) By how many degrees must Owen turn joist 2? Should the 117° angle get bigger or smaller?

Lesson 9: Unit 5 Review

Every rule from the unit on one page, worked examples that mix the topics, and a practice unit test.

1. The Unit on One Page

Use this page to study. Each row is one lesson: what to do, and a short example.

Skill	What to do	Example
Spatial puzzles	Name your strategy: draw or model, make a systematic list, guess and check, eliminate possibilities. **Tangram** (7 pieces): small triangle = 1 unit; square, parallelogram and medium triangle = 2 each; large triangle = 4; the whole square = 16. **Pentominoes**: 12 different shapes made of 5 squares (a flip or a turn is not a new one).	The whole tangram is 80 cm^2. A small triangle is $80 \div 16 = 5$ cm^2.
Spatial games	Play small cases, look for a pattern, and work backward from the end of the game. Take 1, 2 or 3 counters, last counter wins: leave a **multiple of 4**. Tower of Hanoi with n discs: $2^n - 1$ moves.	Pile of 10: take 2, leave 8. 3 discs: $2^3 - 1 = 7$ moves
Measuring and classifying angles	Acute: less than 90°. Right: 90°. Obtuse: between 90° and 180°. Straight: 180°. Reflex: between 180° and 360°. **Protractor**: centre on the vertex, 0 line on one arm, read the scale that starts at 0 on that arm. Reflex angle = 360°− the angle you measure. A polygon with n sides: angle sum = $(n-2) \times 180°$.	Reflex partner of 100°: $360° - 100° = 260°$ Quadrilateral: $(4-2) \times 180° = 360°$
Referents and true bearings	Referents: 22.5°, 30°, 45°, 60°, 90°, 180° (fold a square of paper; a clock is 30° per hour mark). **True bearing**: start at north, turn **clockwise**, write 3 digits. Back bearing: add 180° if the bearing is less than 180°, subtract 180° if it is 180° or more.	Due east = 090° Back bearing of 070°: $070° + 180° = 250°$
Copying and bisecting; perpendicular lines	Copy an angle with a protractor, by tracing or a Mira, or with compass and straightedge. **Bisect**: cut into two equal angles (halve it with a protractor, fold it, or use compass arcs). **Perpendicular** lines meet at 90° (square mark). **Parallel** lines never meet (arrow marks).	Bisect 110°: two angles of $110° \div 2 = 55°$
Angle partners	**Complementary**: add to 90°. **Supplementary**: add to 180° (angles on a straight line). **Vertically opposite** angles are equal. Angles around a point add to 360°. **Adjacent** angles share a vertex and an arm.	40°: complement 50°, supplement 140°
Naming angle pairs	Angles 1–4 at the top crossing and 5–8 at the bottom, clockwise from the upper left (see the picture). Corresponding (F): 1 & 5, 2 & 6, 3 & 7, 4 & 8. Alternate interior (Z): 3 & 5, 4 & 6. Alternate exterior: 1 & 7, 2 & 8. Interior same side (C): 4 & 5, 3 & 6. Exterior same side: 1 & 8, 2 & 7.	
Parallel lines and a transversal	If $\ell_1 \parallel \ell_2$: corresponding, alternate interior and alternate exterior angles are **equal**; interior and exterior angles on the same side **add to 180°**. If the lines are not parallel, these rules do not work. If corresponding angles are equal, the lines **are** parallel.	$\angle 1 = 118°$: $\angle 5 = 118°$ (corresponding), $\angle 2 = 62°$ (supplementary)

Remember. Give a **reason** for every angle you find ("vertically opposite", "corresponding"). Check that the answer fits the picture: an angle that looks acute must be less than 90°. The transversal rules need **parallel** lines (arrow marks or the words "is parallel to"). Bearings: from north, clockwise, 3 digits.

2. Puzzles and Games

Example 1

(a) Two players take turns removing 1, 2 or 3 counters from a pile of 15. Whoever takes the last counter wins. You go first. How many should you take? What do you do after that?

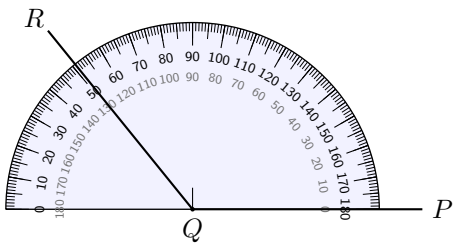
(b) Change the rule: still take 1, 2 or 3, but whoever takes the last counter **loses**. The pile has 18. What should the first player do now?

(c) In the Tower of Hanoi puzzle, you move a stack of discs to another peg, one disc at a time, never putting a bigger disc on a smaller one. Fill in the table. Then find the fewest moves for 6 discs.

Discs	1	2	3	4	5	6
Fewest moves	1	3	7	15		

3. Measuring, Classifying and Estimating Angles

Example 2



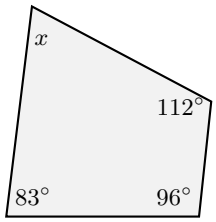
(a) Cover the protractor's numbers and estimate $\angle PQR$ first, using referents.

(b) Measure $\angle PQR$. Which scale did you use, and why?

(c) Classify $\angle PQR$. (d) Find the reflex angle at Q.

Example 3

(a) A loonie has 11 equal sides and 11 equal angles. Find the sum of its angles. Then find the size of each angle, to the nearest tenth of a degree.

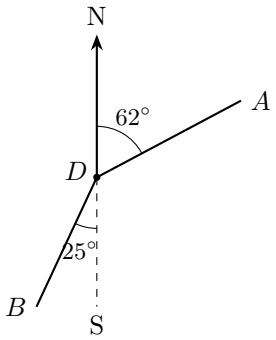


(b) A four-sided building lot has angles of 83°, 96° and 112°. Find the fourth angle, x. (The sketch is not to scale.)

4. True Bearings

Example 4

A float plane is tied up at a dock, D, on a lake in northern Alberta. A is a fishing camp and B is a cabin.



(a) What is the true bearing of the camp, A, from the dock?

(b) What is the true bearing of the cabin, B, from the dock?

(c) The plane flies to the camp. On what bearing does it fly back to the dock?

(d) On what bearing does it fly from the cabin back to the dock?

(e) Sketch a bearing of 300° on the north arrow. Which referent helps?

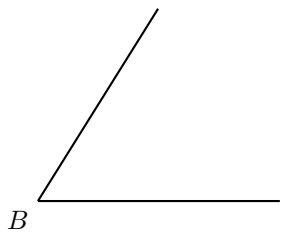


5. Copying and Bisecting Angles; Perpendicular Lines

Example 5

(a) A goalie stands on the line that bisects the angle between the puck and the two goal posts. From the puck, the lines to the two posts make a 14° angle. What angle does the goalie's line make with each post line?

(b) A carpenter must copy the angle where a sloped ceiling meets a wall onto a board. Name two ways to copy the angle.



(c) Bisect this 58° angle with a compass and a straightedge. Write your steps. Check with a protractor.

(d) Sort each pair of lines as parallel, perpendicular or neither. Give a reason.

the two side rails of a ladder

a wall and the floor

the two straight edges of a pizza slice

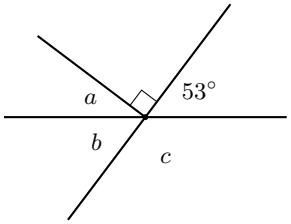
6. Angle Partners

Example 6

(a) Fill in the table. If there is no such angle, write "none" and say why.

Angle	Type	Complement	Supplement
41°			
86°			
111°			

(b) Find a, b and c. Give a reason for each.

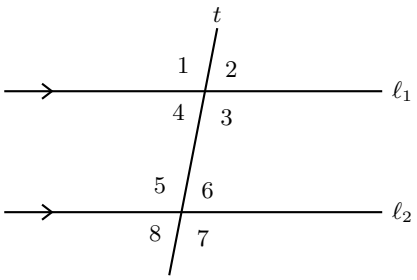


7. Parallel Lines and Transversals

In every numbered diagram, angles 1–4 are at the top crossing and 5–8 at the bottom crossing, going clockwise from the upper left.

Example 7

In the diagram, $\ell_1 \parallel \ell_2$ and $\angle 7 = 101°$.



(a) Name each pair. Then say whether the angles are equal or add to 180°.

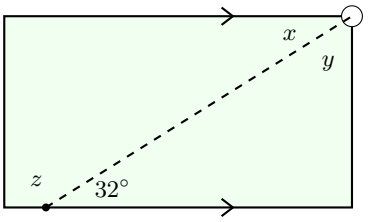
Pair	Name	Equal or 180°?
$\angle 2$, $\angle 6$		
$\angle 3$, $\angle 5$		
$\angle 1$, $\angle 7$		
$\angle 4$, $\angle 5$		
$\angle 1$, $\angle 8$		

(b) Find the other seven angles. Give a reason for each.

Angle	Size	Reason
$\angle 5$		
$\angle 6$		
$\angle 8$		
$\angle 3$		
$\angle 1$		
$\angle 2$		
$\angle 4$		

Example 8

On a pool table, the two long cushions are parallel and the end cushions are perpendicular to them. A ball leaves the bottom cushion at 32° and rolls straight into the top right corner pocket.



(a) Find x, the angle between the ball's path and the top cushion.

(b) Find y, the angle between the path and the end cushion.

(c) Find z.

(d) Two streets are crossed by a diagonal road. At one corner the angle is 86°. At the other street, the corresponding angle is 88°. Are the two streets parallel? Explain.